1. Position / Displacement Sensors

Purpose

Measuring the angle of each robotic arm joint so the controller knows the exact position of the arm.

Sensors Considered

1. Rotary Potentiometer

- Analog output
- Low cost
- Simple to use
- Wears over time
- Medium accuracy

2. Rotary Encoder

- Digital output
- High accuracy
- No mechanical wear
- Better for precise control

3. PSD (Position Sensitive Detector)

- Measures distance using light
- Non-contact
- Not ideal for joint rotation

Selected Sensor: Rotary Encoder

Reason:

Provides higher accuracy and better repeatability than potentiometers. Suitable for closed-loop control of the robotic arm joints.

2. Velocity Sensor

Purpose

A velocity sensor measures the speed of rotation of a motor or joint. It is useful in control systems where speed feedback is required for smooth motion and stability.

Possible Option

Tachogenerator

- Produces a voltage proportional to rotational speed
- Used in traditional motor control systems

Do We Need It?

No, Because the robotic arm uses rotary encoders for position measurement, velocity can be calculated in software:

$$\textbf{Velocity} = \frac{\textbf{Change in position}}{\textbf{Change in time}}$$

The microcontroller reads encoder pulses and computes speed directly.

Replacement Method

Instead of using a separate velocity sensor:

- Velocity is calculated from encoder data.
- This reduces hardware complexity.
- It simplifies wiring and lowers cost.

3. Force / Pressure Sensors

Purpose

To measure the gripping force so the robot does not crush fragile objects like the egg.

Sensors Considered

1. Force Sensitive Resistor (FSR)

- Measures contact force
- Simple analog output
- Easy to connect to Atmega
- Good for light gripping applications

2. Load Cell (Strain Gauge)

- High accuracy
- Used for precise force measurement
- Requires amplifier (more complex)

3. Pressure Sensor (for pneumatic gripper)

- Measures air pressure in cylinder
- Grip force estimated from pressure
- Simple and reliable
- Simple and reliable

Selected Sensor: FSR (Force Sensitive Resistor)

Reason:

Simple, low cost, and sufficient to monitor gripping force and protect fragile objects.

Alternative Sensor Options

In addition to the selected sensors, other sensing technologies were considered for the robotic system.

An **Ultrasonic Sensor** could be used to measure the distance between the gripper and an object or the workspace surface. It operates by transmitting sound waves and measuring the echo return time. This would enable basic obstacle detection and controlled approach movement.

An **Infrared (IR) Sensor** or **Position Sensitive Detector (PSD)** could be used for short-range distance measurement. These sensors provide non-contact detection and can help detect object presence near the gripper.

Limit switches could be installed at the mechanical limits of each joint. They serve as safety devices to prevent over-rotation and mechanical damage, and they can also be used for homing during system initialization.

However, these sensors were not included in the final design in order to maintain simplicity, reduce cost, and minimize system complexity. The selected sensors are sufficient to meet the functional requirements of the robotic arm.